Master's Thesis

1 | Introduction

2 | Methods

In this research, we demonstrate the effects of common class imbalance correction methodologies, when used in the context of clinical prediction model development. We focused on clinical prediction models designed for dichotomous risk prediction. By means of a simulation study, we illustrate the effects of imbalance correction methodologies across 18 unique data-generating scenarios. For each scenario, we compared the out-of-sample predictive performance (i.e., model performance on data not used to train the model) of prediction models developed using a common imbalance corrections in combination with a variety of machine learning algorithms.

When one develops a prediction model in the presence of class imbalance, it is common practice to pre-process data using an imbalance correction technique. Then this pseudo-balanced data is used to train a machine learning algorithm. Each prediction model in our research was developed according to this procedure. We implemented a 5 x 6 factorial design, to compare the out-of-sample predictive performance of prediction models developed with one of 5 imbalance corrections (control and 4 imbalance corrections) and one of 6 machine learning algorithms. In total, we compare 30 prediction model development procedures, each comprised of a unique combination of an imbalance correction and machine learning algorithm.

Anything about ADEMP guidelines for the design and reporting of our simulation study[1] ??

2.1 | Data Generating Scenarios

In our simulation study, 18 (3x2x2) unique data-generating scenarios were selected (*see Table 1*). This was achieved by varying the following three characteristics of the data: the event fraction (proportion of patients with an event), the number of predictors and sample size. Event fraction was varied through the set {0.5, 0.2, 0.02} and number of predictors through the set {8, 16}. In all scenarios, data were generated to yield an expected concordance statistic (C) of 0.85. Given the number of predictors, the event fraction and expected concordance statistic, we computed the minimum required sample size for each prediction model using the R package `pmsampsize`[2]. Sample size was then varied through the set $\{\frac{1}{2}N, N, 2N\}$ implying half the required sample size, adequate samples size and double the required sample size, respectively.

2.2 | Data Generating Mechanism

In this simulation study, we focused on prediction models for dichotomous risk prediction. Consequently, it was necessary to generate data comprised of two classes. We refer to the negative class (non-events) and positive class (events) as class 0 and class 1, respectively. Data were generated such that we could exercise control over the following characteristics of the data: event fraction (proportion of events), number of predictors, sample size and expected concordance statistic.

Data for each class were generated independently using distinct multivariate normal (mvn) distributions. We specified a distinct mean and covariance structure between the classes. As shown in equations 1 and 2,

Scenario	No. Predictors	Sample Size	Event Fraction	n sample	δ_μ	C
1	8	0.5N	0.50	193.00	0.60	0.85
2	8	0.5N	0.20	124.00	0.60	0.85
3	8	0.5N	0.02	899.00	0.60	0.85
4	8	N	0.50	385.00	0.60	0.85
5	8	N	0.20	247.00	0.60	0.85
6	8	N	0.02	1797.00	0.60	0.85
7	8	2N	0.50	770.00	0.60	0.85
8	8	2N	0.20	494.00	0.60	0.85
9	8	2N	0.02	3594.00	0.60	0.85
10	16	0.5N	0.50	193.00	0.49	0.85
11	16	0.5N	0.20	247.00	0.49	0.85
12	16	0.5N	0.02	1797.00	0.49	0.85
13	16	N	0.50	385.00	0.49	0.85
14	16	N	0.20	493.00	0.49	0.85
15	16	N	0.02	3593.00	0.49	0.85
16	16	2N	0.50	770.00	0.49	0.85
17	16	2N	0.20	986.00	0.49	0.85
18	16	2N	0.02	7186.00	0.49	0.85

Table 1: Summary of simulation scenarios.

the mean structure is represented as μ_0 and mu_1 for class 0 and 1, respectively. The covariance matrix is represented as Σ_0 and Σ_1 for class 0 and 1, respectively.

$$\text{Class } \ 0 : \mathbf{X} \sim mvn(\boldsymbol{\mu_0}, \boldsymbol{\Sigma_0}) = mvn(\mathbf{0}, \boldsymbol{\Sigma_0}) \tag{1}$$

$$\text{Class } \ 1 : \mathbf{X} \sim mvn(\boldsymbol{\mu_1}, \boldsymbol{\Sigma_1}) = mvn(\boldsymbol{\Delta_\mu}, \boldsymbol{\Sigma_0} - \boldsymbol{\Delta_\Sigma}) \tag{2}$$

The differences among the parameter values between the two classes are represented in these formulae by $\boldsymbol{\Delta_\mu}$ and $\boldsymbol{\Delta_\Sigma}$; a vector and matrix comprised of the differences in predictor means, and variances/covariances, between the classes, respectively. We specified no variation in means among predictors within a class, making all elements in the vector $\boldsymbol{\Delta_\mu}$ equivalent; denoted by δ_μ. Similarly, we specified no variation in predictor variances within a class, making all diagonal elements in the matrix $\boldsymbol{\Delta_\Sigma}$ equivalent; diagonal elements are denoted by δ_Σ.

For class 0, all predictor means were fixed to zero and all variances to 1. For class 1, all means were non-zero and are represented by the vector $\boldsymbol{\Delta_\mu}$. Finally, in both classes, we allowed 80% of the predictors to covary. All non-zero correlations among predictors in each class were set to 0.2. To ensure the correlation among predictors was not stronger in one class, we fixed the correlation matrices of the two classes to be equal. This was accomplished by computing the off-diagonal elements of $\boldsymbol{\Delta_\Sigma}$ such that the correlation matrices of the two classes were equivalent. Note, the covariance matrices were *not* equivalent between the classes.

For instance, with 8 predictors:

the mean and covariance structure for class 0 is,

$$\boldsymbol{\mu_0} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \boldsymbol{\Sigma_0} = \begin{bmatrix} 1 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0 & 0 \\ 0.2 & 1 & 0.2 & 0.2 & 0.2 & 0.2 & 0 & 0 \\ 0.2 & 0.2 & 1 & 0.2 & 0.2 & 0.2 & 0 & 0 \\ 0.2 & 0.2 & 0.2 & 1 & 0.2 & 0.2 & 0 & 0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 1 & 0.2 & 0 & 0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

and mean and covariance structure for class 1 is,

$$\boldsymbol{\mu_1} = \begin{bmatrix} \delta_\mu \\ \delta_\mu \\ \delta_\mu \\ \delta_\mu \\ \delta_\mu \\ \delta_\mu \\ \delta_\mu \\ \delta_\mu \end{bmatrix}, \boldsymbol{\Sigma_1} = \begin{bmatrix} 1-\delta_\Sigma & z & z & z & z & z & 0 & 0 \\ z & 1-\delta_\Sigma & z & z & z & z & 0 & 0 \\ z & z & 1-\delta_\Sigma & z & z & z & 0 & 0 \\ z & z & z & 1-\delta_\Sigma & z & z & 0 & 0 \\ z & z & z & z & 1-\delta_\Sigma & z & 0 & 0 \\ z & z & z & z & z & 1-\delta_\Sigma & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1-\delta_\Sigma & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1-\delta_\Sigma \end{bmatrix}.$$

Here, $z = (1 - \delta_\Sigma) * 0.2$, to ensure equivalent correlation matrices between the two classes.

2.3 | Scenarios

We simulated data to reflect three unique class imbalance scenarios (Table 1). This was accomplished by varying the event fraction (proportion of patients with the event of interest) through the set {0.5, 0.2, 0.02}. For all scenarios, 8 predictors were generated and sample size (N) was determined as the minimum sample size required for the prediction model (given the number of predictors, event fraction and expected concordance statistic, using the R package `pmsampsize`[2].

For every scenario, parameter values for the data generating distributions (δ_μ and δ_Σ) were selected to generate a concordance statistic (C) of 0.85. Under the assumption of normality for all predictors (in each class), the concordance statistic of the data can be expressed as a function of $\boldsymbol{\Delta_\mu}$, $\boldsymbol{\Sigma_0}$ and $\boldsymbol{\Sigma_1}$[3]. Optimal values of δ_μ and δ_Σ values were computed analytically, based on the following formula[3]:

$$C = \Phi\left(\sqrt{\boldsymbol{\Delta_\mu}' \, (\boldsymbol{\Sigma_0} + \boldsymbol{\Sigma_1})^{-1} \, \boldsymbol{\Delta_\mu}}\right). \tag{3}$$

In equation (1), Φ represents the cumulative density function of the standard normal distribution; $\boldsymbol{\Delta_\mu}$, $\boldsymbol{\Sigma_0}$ and $\boldsymbol{\Sigma_1}$ maintain their previous definitions. To ensure a unique solution, δ_Σ was fixed at 0.3 for each scenario, while equation (1) was solved to yield the appropriate value of δ_μ in each scenario.

Finally, given that data for each class were generated independently, we had direct control over how many observations were generated under each class. The number of observations from the positive class (n_1) was

sampled from the binomial distribution with probability equal to the specified event fraction. The number of observations in the negative class (n_0) was then computed as $N - n_1$, where N is the minimum sample size specified for the prediction model.

Parameter values for the data generating distributions of the simulation scenarios are presented in Table 1.

2.4 | Simulation Methods

Under each simulation scenario, 200 data sets were generated. Each data set was comprised of training and validation data. The training and validation data were generated independently using identical data generating mechanisms. Validation data sets were generated to be ten times larger than the training data sets.

For each generated data set, six prediction models were developed, each using a different classification algorithm. All prediction models were trained using the training data. Out-of-sample performance was then assessed using the validation data.

Classification algorithms were selected based on a systematic review identifying common algorithms used to develop prediction models in a medical context[4]. These algorithms include: logistic regression, support vector machine, random forest and XGBoost. Additionally, based on literature summarizing common strategies to handle class imbalance[? ? ? ?], we included two ensemble learning algorithms designed specifically to handle class imbalance: RUSBoost and EasyEnsemble. Classification algorithms were implemented with their default hyper-parameters; no hyper-parameter tuning was conducted. All classification algorithms and the R packages used for their implementation are summarized in Table 2.

	Name	Abbreviation	R Package
Imbalance Corrections			
	Random Undesampling	RUS	ROSE[5]
	Random Oversampling	ROS	ROSE[5]
	Synthetic Majority Over Sampling	SMOTE	IRIC[6]
	SMOTE - Edited Nearest Neighbours	SENN	IRIC[6]
Classification Algorithms			
	Logistic Regression	LR	base R[7]
	Support Vector Machine	SVM	caret[8]
	Random Forest	RF	caret[8]
	XGBoost	XG	caret[8]
	RUSBoost	RB	ebmc[9]
	EasyEnsemmble	EE	IRIC[6]

Table 2: Summary of imbalance corrections and classification algorithms implemented.

2.5 | Performance Metrics

Out-of-sample model performance was assessed using measures of calibration, discrimination and overall performance. All performance metrics were computed using validation data.

Calibration was measured using visual and empirical metrics. For each simulation iteration, we fitted a flexible calibration curve for each model; this was done with loess regression using `ggplot2`[10]. In a flexible calibration curve, when estimated probabilities (x-axis) correspond well with the observed proportions in the

data (y-axis), the curve follows a diagonal line ($y = x$)[11]. In addition to the calibration plots, calibration intercept and slope were calculated. With respect to calibration intercept and slope, ideal calibration is represented by values of 0 and 1, respectively[12].

Discrimination refers to a model's ability to distinguish between the classes[?] . The concordance statistic was used to measure model discrimination; computed using the R package `pROC`[13]. For dichotomous risk prediction, it is equivalent to the area under the Receiver Operator Characteristic curve[? 12]. A model which perfectly discriminates between the classes will have a concordance statistic of 1; the minimum value for this statistic is 0.5[12].

Overall performance was measured by the Brier score. This metric reflects both model discrimination and calibration and is calculated according to the following formula[12]:

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}(p_i - o_i)^2, \tag{4}$$

where N is the sample size, p_i represents the estimated probability for the ith individual and o_i represents the observed outcome (0 or 1) for the ith individual. In an ideal model, estimated probabilities approximate the observed outcome well for all individuals; ideal models produce a Brier score near to zero.

For empirical measures of model performance (concordance statistic, Brier score, calibration intercept and calibration slope), the mean over all iterations and corresponding Monte Carlo error were reported.

3 | Results

4 | Disucssion

References

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